

Curtin University, Semester 2, 2026
ECOM5005
Tutorial 1

Remember that you are expected to prepare for the tutorial *prior* to your tutorial. In particular, if you cannot obtain answers then you should prepare your own questions concerning these problems so you can learn. Most of the time class will be devoted to the short answer questions, however some time towards the end of the tutorial will be reserved for the data exercise (Part 3) and MyLab (Part 1) for any students who require assistance with the technical computing aspect.

Part 1 – Pearson MyLab

Please logon to MyLab: <https://www.pearsonmylabandmastering.com/au/>

Note: Please refer to the instructions in the welcome module to obtain access to MyLab. The instructions also include various options to purchase the essential book for this unit.

Once you are logged in, please attempt the following activities:

- [1] Complete the candidate agreement – This is a declaration that all students will complete prior to sitting an online assessment. We are doing this in week 1 so that students are familiar with this setting.
- [2] Complete Homework #1 – These are a set of simple questions designed to assist students to get started on the MyLab platform.

Note: If you are having issues with obtaining access, please re-read the instructions. If it is still not working out, please send an email to help@pearson.com.au explaining the issue. You may complete the above exercises once you have access.

Part 2 – Short answer questions

Question 1.

‘A survey indicates that the vast majority of university students own their own personal computer.’ What information would you want to know before you accepted the results of this survey?

Question 2.

The online restaurant search site Dimmi <www.dimmi.com.au> encourages diners to rate restaurants they have been to by giving them reward points which can be accumulated until a meal discount is available. A restaurant at The Rocks in Sydney has been rated as follow: Recommended 8.7; Food 8.5; Service 8.7; Value for money 7.8; Atmosphere 8.4. Why might you expect these ratings to differ from those obtained if all diners were to rate the restaurant?

Question 3.

An ECOM 5005 tutorial has 40 students enrolled and 17 of these are male. Hence the proportion of students that are male is $17/40 = 42.5\%$. Consider the statement “a majority of student are female”. In considering whether this statement is valid or not, assume the values of 17 and 40 are correct but consider the meaning of the key words in the statement (e.g. “majority”, “student” and “female”). Consider alternative meanings, or at least be prepared to discuss alternative meanings from other students during your tutorial.

Question 4 (thinking of a possible project for ECOM5005)

Consider an interesting situation where evidence to answer a question must realistically be obtained from a subset or sample of the population of interest. Try to think of an example that is non-trivial and perhaps related to your work situation. A simple example would be investigating “what proportion of employees (or of the general population) are willing to have a Covid vaccination?” or “What proportion of employees (or of the general population) believe vaccination should be compulsory for everyone?” A more complicated version might investigate whether this proportion differs between men and women (or age or some other characteristic).

Part 3 – Data exercise

Note: this is a gentle introduction to Excel to make sure everyone can do some basic operations we need this semester. Some students will find this easy. Some students will find it harder, or even intimidating. If this is you, don’t panic – there will be time in the first tutorial class to get help. It is important you try first so we know where you need help.

- First, open the link titled “survey questions” and the link titled “survey results” for Topic 1: Data. The latter contains answers provided by 62 people and the former contains the questions.

Below are instructions specific to our data, however there are better ways to find out how to do something. For example, google “how to make a histogram in excel”. You will probably get over 3 million pages! Some are better than others, but the Microsoft one starting “support.microsoft.com...” is not bad (it was written by the owners of Excel, but other people have created videos if you prefer that).

It is up to you how much time you spend learning to do things (and making them “pretty”) in Excel. The procedures below are not necessarily the best or easiest, so use your own way (or google).

In this tutorial we will investigate the WAM (weighted average mark) of the males and females. The below instructions are designed to produce the following.

- Histograms of the WAM values, preferably for males and females separately, and preferably using the same scale on the horizontal axes.
- Descriptive statistics for the WAM values, preferably for females and males separately.

If you know how to do this, or happy to use google to teach yourself how to do it, then ignore the instructions below. Answers (Histograms and descriptive statistics) are at the end of this tutorial.

- 1) Open the data “survey results”.
- 2) I suggest you use the “File – Save As” option to save the file in a new name “tut1.docx” so you still have the original data in its original file after you save your work (remember to save the file before you exit Excel).
- 3) Sort the data so the males or females are at the top or bottom of the sheet. Do this by
 - a. Selecting (this means clicking with the left mouse button) all the data (left mouse clicking on the triangle above row 1 and to the left of column A).
 - b. Selecting the “Data” menu at the top and then selecting “Sort” (which has a box with the letters A and Z in it).
 - c. In the “Sort by” option, click (left mouse button) on the arrow and select “Gender”. You can then click “OK” at the bottom. There are other options here, but the default is to sort in alphabetical order (so you will see females come before males in your data).
- 4) Highlight the WAM for the females (do this by left clicking on the first WAM, cell F2, and then “drag the mouse” (this means hold the right mouse button down while you move the mouse) down to the last female WAM, cell F34, before releasing pressure on the mouse button).
- 5) Select the “Insert” tab at the top of Excel and then “Recommended charts”. You will see a chart of the data, with alternative options on the left. When I did this I saw on the right a “Clustered Column” chart, which has 33 rectangles and the height of each rectangle is the WAM of the student (the first one is between 60 and 70, and when I look at my data I see the WAM of the first female is 64.4).
 - a. You might wish to click “OK”, in which case you will get this “Column Chart” appearing in your Excel data. I did not bother because apart from showing me what all the data looks like, I do not find this option very useful. If you do click OK, repeat the above instructions so you can get back to this window before proceeding with the next step.
 - b. Look further down the list on the left (you may need to drag the grey rectangle down) until you find a chart with only about 5 rectangles (the first one you find like this). Click on this picture and the figure on the right should change into a “Histogram”. Click OK.
 - c. You can change the title from “Chart Title” to something else (like “Histogram of female WAM”) by clicking in the Chart Title and typing your title and deleting their title).
- 6) Repeat the previous step, but this time highlight the WAM for the males (rows 35 to 62) and instead of selecting “Recommended Charts”, select the picture of the histogram (the all blue rectangles) and then select the histogram (top left picture). You will now find another histogram for male WAM values. Look carefully at the values on the horizontal axis – why is it difficult for your eye to quickly tell whether males or females tend to have the highest WAM?
- 7) We will add in some functions that are useful for this unit. Click on the File menu, then select “Options” (bottom left), then click “Add-in” near the bottom left. Select “Analysis ToolPak” near the top and click OK. It may appear as if nothing has

happened, but we will see below something has. We will use this often, so it is important you successfully do this step!

- 8) Calculate some numerical summaries of the female WAM values by
 - a. Click the Data tab at the top of Excel, and then click on “Data Analysis” on the far right near the top (this was not there before you did the Add-in in the step above).
 - b. Select “Descriptive Statistics” (the options are in alphabetical order – we will use many of these options later).
 - c. Click on the Input range box, and then highlight the cells F2 to F34 (or type in F2:F34). Also click on the “Summary statistics” box towards the bottom and then click OK. You will see a new tab appears (see the bottom of your Excel window) and in this will be lots of numbers. The top one is the Mean (or average) value of (about) 64.7. You may or may not know what any of these are (we will learn about the important ones so do not worry). At the bottom you will see the count is 33 (there are 33 females) and the lowest (Minimum) WAM is 27.9 and the highest (Maximum) WAM is 93.9 (for the females).
- 9) Repeat the previous step but this time for the male WAM values.
- 10) It is much more useful to have histograms that use the same ranges for males and females. This can be done using the “Histogram” option instead of the Descriptive statistics option in “Data Analysis” (see above steps). Follow the steps:
 - a. Type the number 10 in one cell and the number 20 in the cell beneath it. You may wish to place this in a separate tab to the data (you can create a new tab by clicking on the “+” at the bottom of the Excel window).
 - b. Highlight these two cells (click on the 10 and then drag the mouse to the second value before releasing the left mouse button). If you look carefully at these two highlighted cells, the bottom right should be a little square (a small dot). Select this and drag it down about 8 cells. You should see numbers appear to continue the sequence, from the 10 and 20 to include 30, 40, etc. We want this to go down to 100 (theoretically, the maximum WAM).
 - c. Now use the Data menu and select “Data Analysis” again. This time select “Histogram”. For the “Input Range”, select the female WAM values again. For the “Bin range”, select the cells with your numbers 10 to 100 (these may all be on separate sheets, so you need to use the tabs at the bottom of the Excel window to move around). Finally, either click on a cell for the “Output range” (I suggest to the right of your values 10 to 100). You should see your bin values repeated and also the number of people in each bin. For example, there are 11 females in the bin called 70 (so there are 11 females with a WAM greater than 60 and less than or equal to 70).
 - d. Repeat the previous step to get values for the males. I suggest you choose a different location for the output – perhaps a few cells to the right of the output for the females.
 - e. The numbers you produce have useful information, but we can create a prettier picture in the form of a histogram. One way to do this is with the following steps:
 - i. Select the cells with your “Bin” values 10 to 100. Now push the “Ctrl” key on your keyboard but keep it held down (using a finger on your left hand) while you use your mouse to select the “Frequency” values for the females. Make sure you select the same number of cells as the number of Bin cells you selected. Repeat this process (using the “Ctrl” key again) to select the

male Frequencies. Once you have these 3 groups of numbers selected you can release the “Ctrl” key.

- ii. Select “Insert” and then “Recommended Charts”. The second chart is a “Clustered Column”. You should see rectangles in two colours (one is females and one is males), with the horizontal axis being the maximum bin range and the vertical axis being the number of people. For example, the bin “60” (meaning WAM greater than 50 and less than or equal to 60) has 9 females and 8 males.
- iii. You can change some of the characteristics of this figure by clicking (or double clicking with the right mouse button) the part you wish to change. For example, you can change the title. We give two simple examples next.
- iv. The colours of the rectangles can be changed by double clicking on one of the rectangles. Experiment with the options and you can change the colours and shading of the rectangle and its boundary (sometimes useful, depending on your artistic preferences).
- v. The labels of the data (default is “series 1” and “series 2”) can be changed by clicking the right mouse button somewhere in the figure and then selecting “Select Data” (using the left mouse button). On the left, you will see the series names. Click on “series 1” and type “females” into the “Series name” box. Click OK. Repeat for series 2 for the males. When you are happy with the new labels exit this window by selecting OK again.

